

Orpheus:

A New Deep Learning Framework for Easy Deployment and Evaluation of Edge Inference

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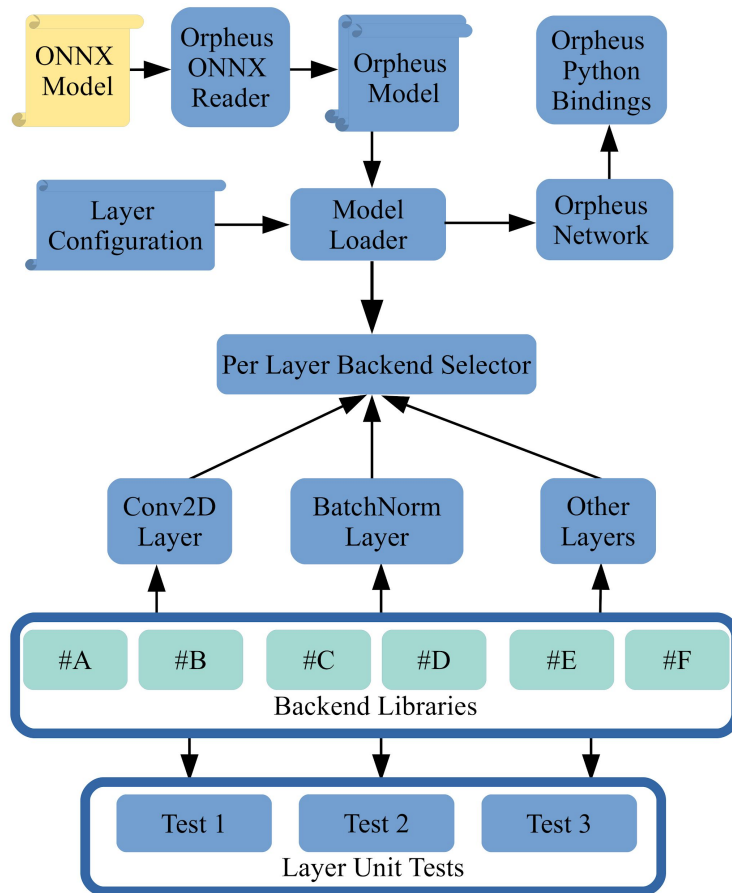
Introduction

- Existing **deep learning frameworks** have various design trade-offs
- Often support full suite of **features** such as training and deployment
- **Debugging** performance can be difficult due to complexity of codebases
- **Inference optimisations** often hidden from user (e.g. algorithmic primitives)

This work looks at how frameworks performing neural network inference can support systems research, and introduces **Orpheus - a new inference framework for neural networks**

Orpheus Design

- **Multiple backends** for layers
- Can import **ONNX models**
- **Python bindings** for easy data collection
- Encourages user extensions and **experimentation**



DNN Framework Features

- **Low-level modifications:** Ability to access lower-level features of the platform
 - e.g. SIMD intrinsics
- **Model interoperability:** Support for models trained in other frameworks
 - e.g. via the ONNX format
- **Platform compatibility:** Ease of deployment on a range of edge devices
 - e.g. different HW features
- **Codebase accessibility:** Ability to prototype/test/integrate new features or backends
- **Performance (inference time):** Execution time to evaluate data on common NNs
 - e.g. classify images

Existing frameworks

- **TensorFlow Lite (TF-Lite)**
 - Subset of full TensorFlow operations
 - Works with few dependencies across a range of platforms
 - Inference only
- **PyTorch**
 - Training and inference
 - High-level API, good for prototyping
 - Abstractions create barriers for investigating performance issues
- **DarkNet**
 - Small, simple codebase
 - Lacks competitive performance
 - Poor third party model support
- **TVM**
 - Deep learning compiler stack
 - Competitive performance
 - Integration of new features can have high development overheads

Orpheus

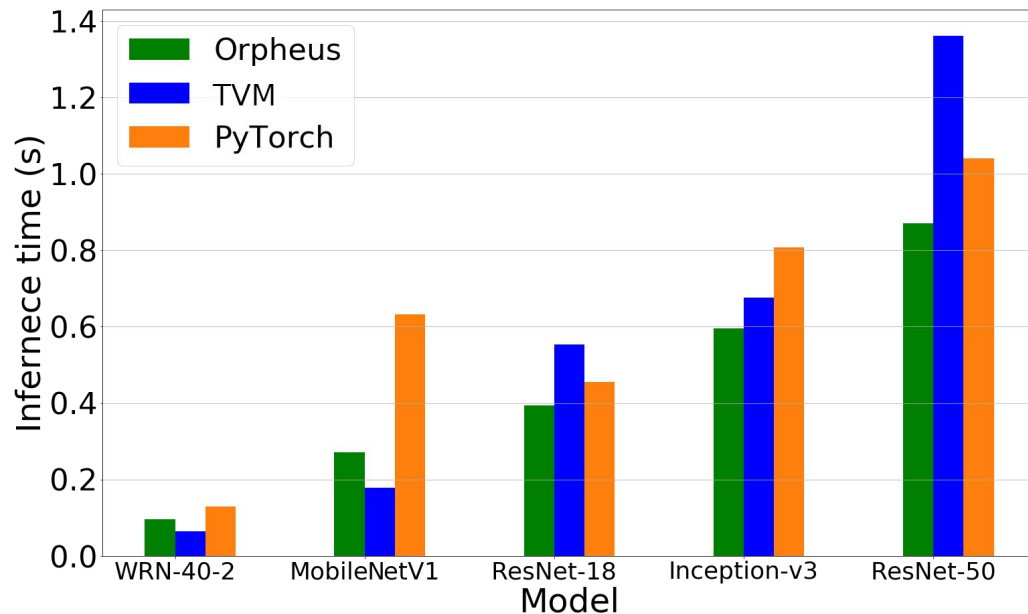
- Values a **small codebase**
- **Easy integration** of lower level libraries
- **Transparent** choice of backend for each network layer
- Imports models from popular framework interoperable **ONNX format**

Frameworks comparison

Comparison of frameworks based on our experience, rated 1-3

Feature	Framework				
	TF-Lite	PyTorch	DarkNet	TVM	Orpheus
Low-level modifications	1	1	2	2	3
Model interoperability	2	3	1	3	3
Platform compatibility	3	2	3	3	3
Codebase accessibility	1	2	3	1	3
Performance (inference time)	2	2	1	2	3

Evaluation



1.4x faster than *PyTorch*

1.2x faster than *TVM*

7x faster than *Darknet*

- Inference time in seconds, on a single big CPU core of the Hikey 970
- **DarkNet** times too high, **TF-Lite** via Python did not permit single thread inference

Conclusions and Future Work

- Orpheus is a **new deep learning framework** with good properties for systems research
- It can achieve competitive performance on some benchmarks
 - **1.4x, 1.2x, 7x** for PyTorch, TVM, and DarkNet respectively
- Extendible to new backends, with a modular design
- **Future work:**
 - Add additional third party backends to Orpheus
 - Cover full set of ONNX operations

Thank You!

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